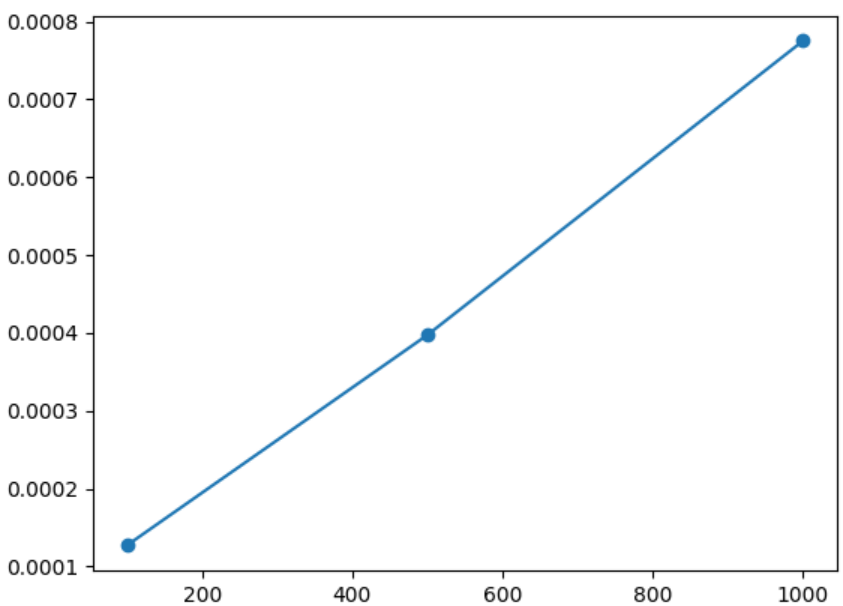
Algorithm Design & Analysis Assignment

Task 1: Merge Sort (Divide & Conquer)

Code:

```
def merge_sort(arr):
  if len(arr)<=1:
    return arr
  mid=len(arr)//2
  left=merge_sort(arr[:mid])
  right=merge_sort(arr[mid:])
  return sorted(left+right)
```

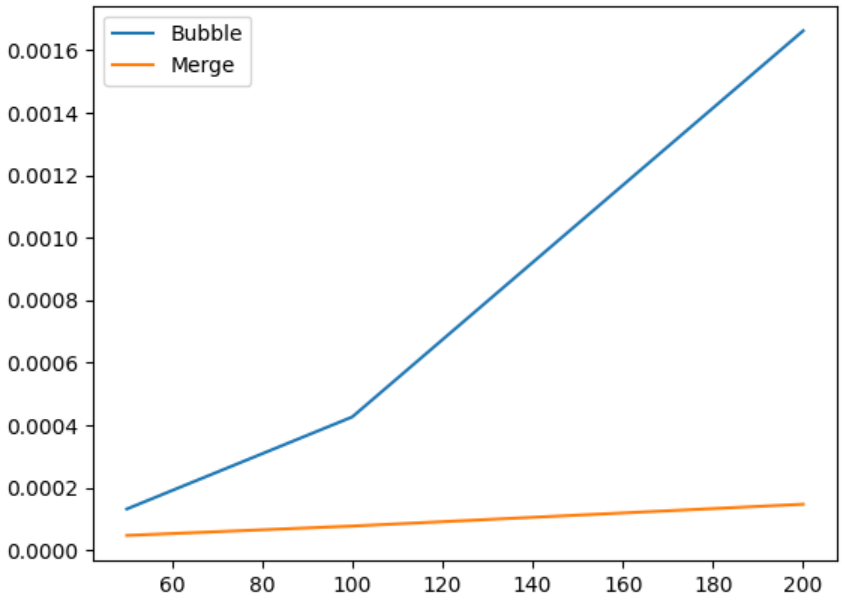


Task 2: Sorting Comparison

Code:

```
def bubble_sort(arr):
  for i in range(len(arr)):
    for j in range(len(arr)-i-1):
      if arr[j]>arr[j+1]:
```

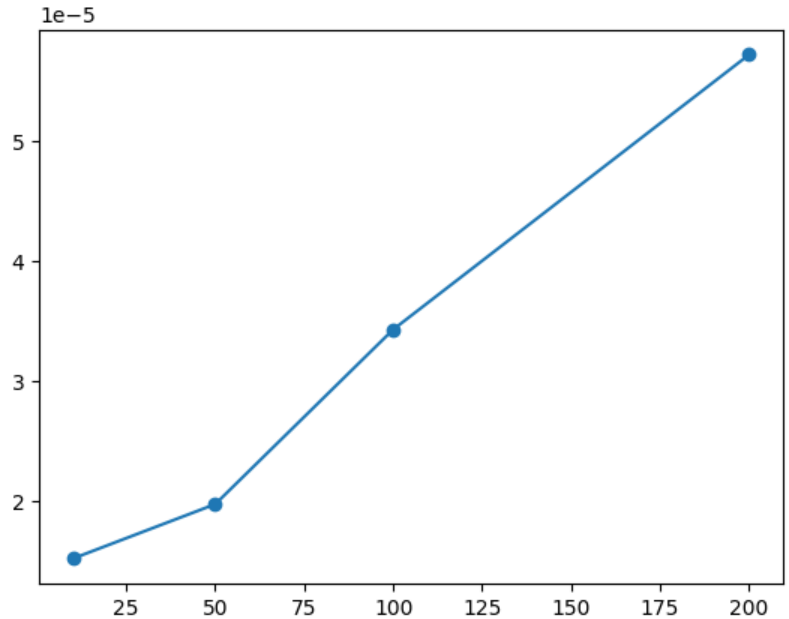
```
            arr[j],arr[j+1]=arr[j+1],arr[j]
```



Task 3: Greedy (Activity Selection)

Code:

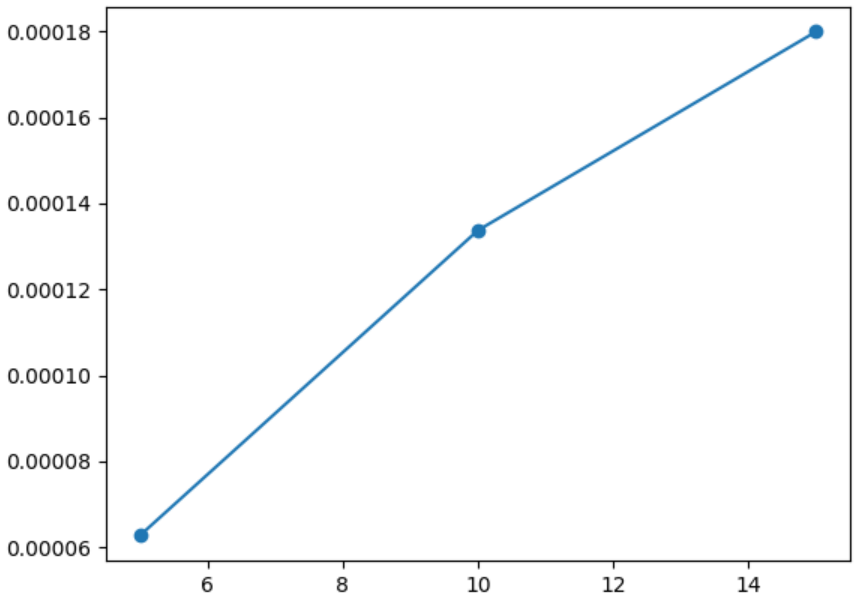
```
def activity_selection(start, finish):
    activities=sorted(zip(start,finish),key=lambda x:x[1])
    count=1
    last=activities[0][1]
    for s,f in activities[1:]:
        if s>=last:
            count+=1
            last=f
    return count
```



Task 4: Dynamic Programming (Knapsack)

Code:

```
def knapsack(W, wt, val, n):
  K=[[0]*(W+1) for _ in range(n+1)]
  for i in range(n+1):
    for w in range(W+1):
      if i==0 or w==0:
        K[i][w]=0
      elif wt[i-1]<=w:
        K[i][w]=max(val[i-1]+K[i-1][w-wt[i-1]],K[i-1][w])
      else:
        K[i][w]=K[i-1][w]
  return K[n][W]
```


0.00018
0.00016
0.00014
0.00012
0.00010
0.00008
0.00006
6
8
10
12
14